

EDA Team 1: Presentation 2

Meera, Sana, Adam, Marcos

Objective: Call Abandonment Scenarios

Established Examples for

1. Customer left after long wait time
2. Customer left after calling during closed hours
3. Customer left after tenant deterrence menu

Inferred Examples for

1. Customer left while hearing something play (ex: music)
2. Customer left after/during the courtesy callback
3. Customer left after reaching farmworker menu

Customer Left after Long Wait Time

On FarmworkerMainMenu and left after 24 minutes.

calls_customer_left[calls_customer_left["Call ID"] == 4]

	Call ID	Contact Session ID	EP Name	Flow Name	Activity Name	Queue Name	Termination Reason	Activity Start Timestamp	Time Difference	weekday
31	4	6364de29-c942-4e6f-95fe-4a63ee631450	Farmworker Main Number Telephony EP	NaN	NaN	NaN	NaN	2025-03-16 07:27:28	NaN	Sunday
32	4	6364de29-c942-4e6f-95fe-4a63ee631450	NaN	FarmworkerMain	NaN	NaN	NaN	2025-03-16 07:27:28	0.0	Sunday
33	4	6364de29-c942-4e6f-95fe-4a63ee631450	Farmworker Main Number Telephony EP	NaN	LanguageSelectionMenu	NaN	NaN	2025-03-16 07:27:28	0.0	Sunday
34	4	6364de29-c942-4e6f-95fe-4a63ee631450	Farmworker Main Number Telephony EP	FarmworkerMain	NaN	NaN	NaN	2025-03-16 07:27:28	0.0	Sunday
35	4	6364de29-c942-4e6f-95fe-4a63ee631450	Farmworker Main Number Telephony EP	NaN	FarmworkerMainMenu	NaN	NaN	2025-03-16 07:27:36	8.0	Sunday
36	4	6364de29-c942-4e6f-95fe-4a63ee631450	Farmworker Main Number Telephony EP	NaN	NaN	NaN	Customer Left	2025-03-16 07:51:33	1437.0	Sunday

Customer Left after calling during Closed Holiday Hours

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[94]: calls_customer_left[calls_customer_left["Call ID"] == 3]
```

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[94]:
```

Call ID	Contact Session ID	EP Name	Flow Name	Activity Name	Queue Name	Termination Reason	Activity Start Timestamp	Time Difference	weekday
11	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Main Number Telephony EP	NaN	NaN	NaN	2025-03-16 02:39:46	NaN	Sunday
12	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	NaN	LACMain	NaN	NaN	2025-03-16 02:39:46	0.0	Sunday
13	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Main Number Telephony EP	NaN	LanguageSelectionMenu	NaN	2025-03-16 02:39:46	0.0	Sunday
14	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Main Number Telephony EP	LACMain	NaN	NaN	2025-03-16 02:39:46	0.0	Sunday
15	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	NaN	ClosedHoursHolidaysMenu	NaN	NaN	2025-03-16 02:39:56	10.0	Sunday
16	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Closed Hours-Holidays Menu Telephony EP	NaN	ClosedMenu	NaN	2025-03-16 02:39:56	0.0	Sunday
17	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Closed Hours-Holidays Menu Telephony EP	NaN	AddressFaxHoursMenu	NaN	2025-03-16 02:40:48	52.0	Sunday
18	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Closed Hours-Holidays Menu Telephony EP	NaN	ClosedMenu	NaN	2025-03-16 02:41:21	33.0	Sunday
19	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Closed Hours-Holidays Menu Telephony EP	NaN	StaffDirectoryEnglishTransfer	NaN	2025-03-16 02:41:30	9.0	Sunday
20	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	NaN	NaN	Staff Directory English Transfer	NaN	2025-03-16 02:41:30	0.0	Sunday
21	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	NaN	2025-03-16 02:41:30	0.0	Sunday
22	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	NaN	2025-03-16 02:41:30	0.0	Sunday
23	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	NaN	2025-03-16 02:41:30	0.0	Sunday
24	3	1b35074f-6d8b-4132-99a2-32ae83ea54cf	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	Customer Left	2025-03-16 02:42:18	48.0	Sunday

Staff Directory English Transfer	NaN	2025-03-16 02:41:30	0.0	Sunday
Staff Directory English Transfer	NaN	2025-03-16 02:41:30	0.0	Sunday
Staff Directory English Transfer	Customer Left	2025-03-16 02:42:18	48.0	Sunday

Customer left during closed hours, but also during a Staff Transfer?

Why are there three rows for the transfer and 48 seconds of time when the transfer recording is only 5 seconds long?

Customer left during Courtesy Callback?

Rows 10 and 255: Customer Left during Courtesy Callback Telephone EP.

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calls_customer_left[
  (calls_customer_left["Call ID"] == 2) &
  (calls_customer_left["weekday"] == "Sunday")
]
```

	Call ID	Contact Session ID	EP Name	Flow Name	Activity Name	Queue Name	Termination Reason	Activity Start Timestamp	Time Difference	weekday
5	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	NaN	NaN	NaN	Consumer SP	Queue Timeout	2025-03-16 02:15:16	NaN	Sunday
6	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	NaN	NaN	NaN	Consumer SP	Customer Left	2025-03-16 02:15:16	0.0	Sunday
7	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	Courtesy Callback Telephony EP	NaN	CallbackRetry	NaN	NaN	2025-03-16 02:15:26	10.0	Sunday
8	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	NaN	NaN	NaN	Consumer SP	NaN	2025-03-16 02:15:26	0.0	Sunday
9	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	NaN	NaN	NaN	Consumer SP	Customer Left	2025-03-16 02:15:26	0.0	Sunday
10	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	Courtesy Callback Telephony EP	NaN	NaN	Consumer SP	Customer Left	2025-03-16 02:15:26	0.0	Sunday
246	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	NaN	NaN	NaN	Consumer SP	Customer Left	2025-03-16 23:15:26	75600.0	Sunday
247	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	NaN	NaN	NaN	Consumer SP	Queue Timeout	2025-03-16 23:15:26	0.0	Sunday
252	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	Courtesy Callback Telephony EP	NaN	CallbackRetry	NaN	NaN	2025-03-16 23:15:36	10.0	Sunday
253	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	NaN	NaN	NaN	Consumer SP	NaN	2025-03-16 23:15:36	0.0	Sunday
254	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	NaN	NaN	NaN	Consumer SP	Customer Left	2025-03-16 23:15:36	0.0	Sunday
255	2	7cb76bad-f177-4fdb-b0fd-681f9b2cceb0	Courtesy Callback Telephony EP	NaN	NaN	Consumer SP	Customer Left	2025-03-16 23:15:36	0.0	Sunday

Customer left and courtesy callback in 10 seconds→ customer got cut off

Customer left after staying in the queue for too long.

25390a18-15db-4e03-b5f2-bdc52d856154	PreQueueMessage2	NaN	2025-03-17 08:04:19	0.0
25390a18-15db-4e03-b5f2-bdc52d856154	PlayMOH300s	NaN	2025-03-17 08:04:31	12.0
25390a18-15db-4e03-b5f2-bdc52d856154	QueueMenu1	NaN	2025-03-17 08:09:33	302.0
25390a18-15db-4e03-b5f2-bdc52d856154	PlayMOH300s	NaN	2025-03-17 08:09:54	21.0
25390a18-15db-4e03-b5f2-bdc52d856154	QueueMenu1	NaN	2025-03-17 08:14:56	302.0
25390a18-15db-4e03-b5f2-bdc52d856154	PlayMOH300s	NaN	2025-03-17 08:15:17	21.0
25390a18-15db-4e03-b5f2-bdc52d856154	QueueMenu1	NaN	2025-03-17 08:20:19	302.0
25390a18-15db-4e03-b5f2-bdc52d856154	PlayMOH300s	NaN	2025-03-17 08:20:41	22.0
25390a18-15db-4e03-b5f2-bdc52d856154	NaN	Customer Left	2025-03-17 08:25:25	284.0

- PlayMOH300s refers to the 300 second long waiting music
- QueueMenu1 refers to the repeating dialogue during the queue between the music (asks client to give a callback number)
- From calls in which customers left during a queue, their calls lasted on average **255 seconds** → When is a closed queue decided versus maintaining queues?

Customer leaves after closed queue menu:

7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	LegalMenu1	NaN	2025-03-17 08:34:47	0.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	LegalMenu2	NaN	2025-03-17 08:35:23	36.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	LegalMenu2	NaN	2025-03-17 08:37:30	127.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	LegalMenu2	NaN	2025-03-17 08:37:34	4.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	NaN	NaN	2025-03-17 08:38:32	58.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	HousingMenu	NaN	2025-03-17 08:38:32	0.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	PreTenantMenu	NaN	2025-03-17 08:38:53	21.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	TenantMenu	NaN	2025-03-17 08:39:36	43.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	NaN	NaN	2025-03-17 08:39:51	15.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	NaN	NaN	2025-03-17 08:39:51	0.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	ClosedQueueMenu	NaN	2025-03-17 08:39:51	0.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	ClosedQueueMenu	NaN	2025-03-17 08:40:07	16.0
7290128c-3ff7-4d1f-a5b0-dc8bd24b1a68	NaN	Customer Left	2025-03-17 08:40:07	0.0

This is the second half of one customer's call journey. After going through several menus (and previously the seniors menus and submenus) the customer left after being told the queue was closed. → How can closed queue menus be brought up earlier in the menu, if at all?

Customers leaving after the Tenant Deterrence Menu

In this scenario, the customer left after receiving the tenant deterrence dialogue and going back to the tenant menu. Most calls like this were found to have the customer return to the tenant menu after the deterrence menu, then leaving the call. → The deterrence menu is not providing enough information for customers to seek assistance elsewhere.

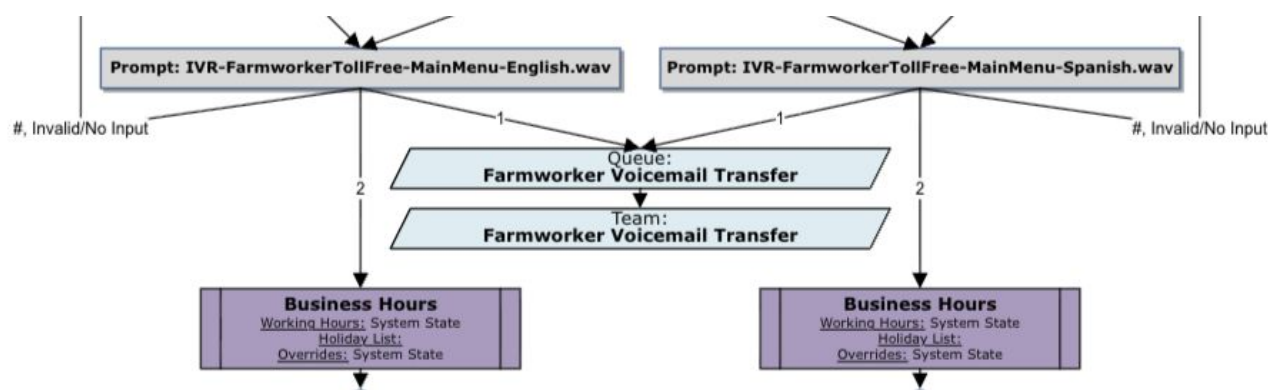
43fea056-10d1-4a96-8d3d-eeaf83cc9906	HousingMenu	NaN	2025-04-02 16:02:36	0.0
43fea056-10d1-4a96-8d3d-eeaf83cc9906	PreTenantMenu	NaN	2025-04-02 16:02:59	23.0
43fea056-10d1-4a96-8d3d-eeaf83cc9906	TenantMenu	NaN	2025-04-02 16:03:41	42.0
43fea056-10d1-4a96-8d3d-eeaf83cc9906	TenantDeterrenceMenu	NaN	2025-04-02 16:04:12	31.0
43fea056-10d1-4a96-8d3d-eeaf83cc9906	TenantMenu	NaN	2025-04-02 16:06:26	134.0
43fea056-10d1-4a96-8d3d-eeaf83cc9906	NaN	Customer Left	2025-04-02 16:06:26	0.0

Unfortunately, we have limited resources and are only able to take certain types of housing cases. The following resources may be helpful to you. To get free legal advice and referrals, please call CARPLS at 312-738-9200. If you have internet access, please visit illinoislegalaid.org. Chicago tenants can also get guidance to resolve issues with your apartment at squaredawaychicago.com. Finally, you may be able to meet with a free lawyer to get advice about your legal issue at one of our monthly neighborhood advice clinics.

For information about upcoming clinic dates and locations, press 1. To repeat this message, press the pound key. To return to the previous menu, press 9.

Customers leaving after the Farmworkers Menu

A common occurrence is the customer waiting for a long time after selecting the farmworker menu and then leaving.



You have reached the Illinois Farmworker Legal Services Project. If you work in agriculture, including landscaping, work related to the production of food, plants or care of animals, press 1. Otherwise, press 2. To repeat these options, press the pound key.

99a6c025-3417-4f0a-a952-c42f4756a712		NaN	NaN	2025-07-01 17:43:57	NaN
99a6c025-3417-4f0a-a952-c42f4756a712		NaN	NaN	2025-07-01 17:43:57	0.0
99a6c025-3417-4f0a-a952-c42f4756a712	LanguageSelectionMenu		NaN	2025-07-01 17:43:57	0.0
99a6c025-3417-4f0a-a952-c42f4756a712		NaN	NaN	2025-07-01 17:43:57	0.0
99a6c025-3417-4f0a-a952-c42f4756a712	FarmworkerMainMenu		NaN	2025-07-01 17:44:04	7.0
99a6c025-3417-4f0a-a952-c42f4756a712		NaN	Customer Left	2025-07-01 18:12:13	1689.0

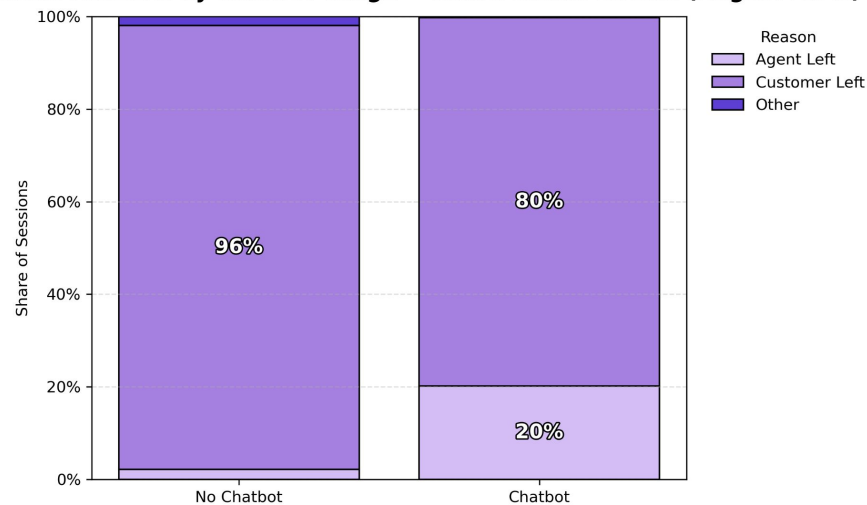
Termination Reason Analysis

The figure describes the termination reasons of calls within the Family Menu during August (by calls that reached the chatbot vs ones that did not)

Takeaway: Calls that reached the chatbot seemed to have a **more positive outcome** due to the higher proportion of agent ended calls

Next Step: Limit control group to calls that reached a human agent or compare termination reasons during the broad chatbot period vs before/after the chatbot period

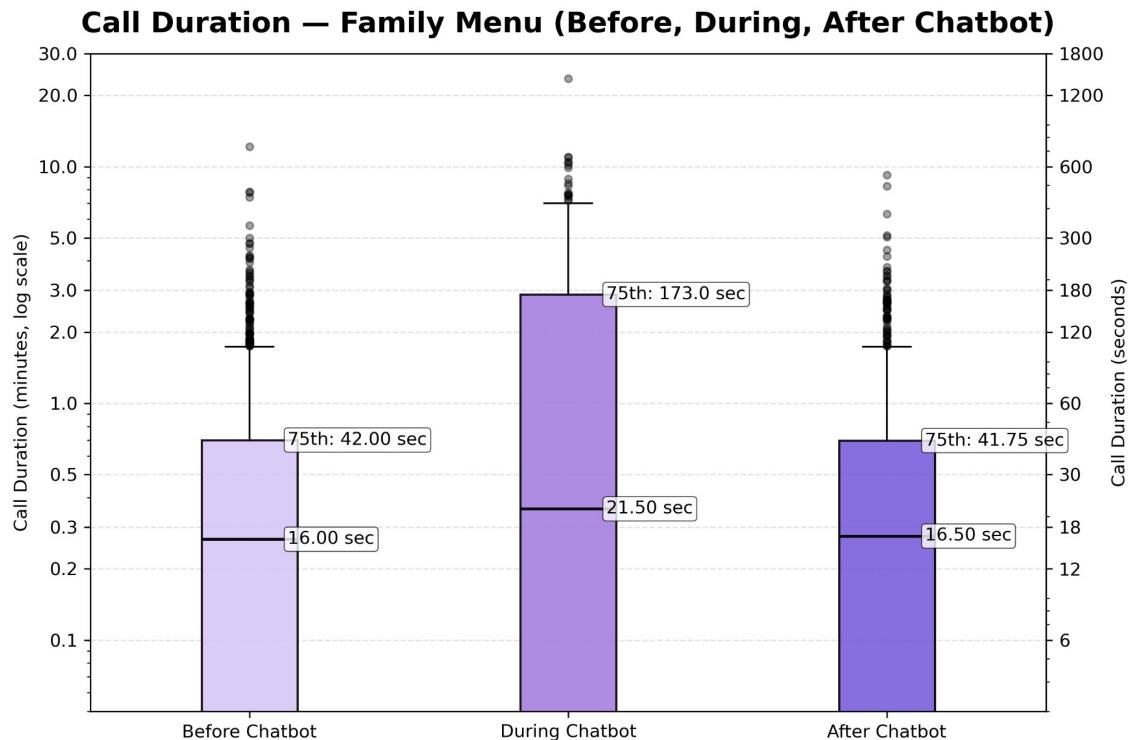
Termination Reasons by Chatbot Usage — Share within Bucket (August 2025)



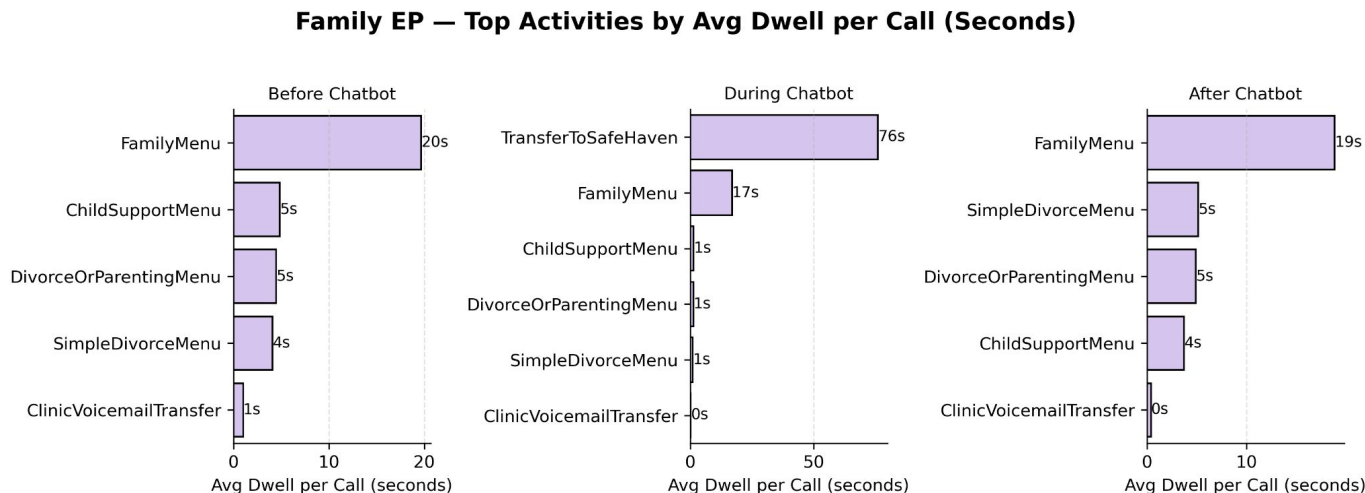
Call Duration Analysis

The figure describes the distribution of call duration within the family menu before/after and during the Chatbot usage period

Takeaway: Calls during the Chatbot's usage seem to be longer and have a wider range in length



Time Spent within Family Menu



The figure describes the average time spent per call in different activities within the family menu

Takeaway: A majority of the time spent during the chatbot usage period was with the chatbot. There was also much less time spent within submenu's within the family menu during the chatbot usage period